B2 Algebra and quadratics: solutions

← Module overview · Problems

Try each problem seriously before you read its solution. When your answer differs, find the exact line where your working first goes wrong.

Practice

1.

(a) $3x(2x-5) = 6x^2 - 15x$

(b) $(x+4)(x-7) = x^2 - 7x + 4x - 28 = x^2 - 3x - 28$

(c) $(2x-3)^2 = (2x)^2 - 2 \cdot 2x \cdot 3 + 3^2 = 4x^2 - 12x + 9$

(d) $(5a+2b)(5a-2b) = (5a)^2 - (2b)^2 = 25a^2 - 4b^2$

(e) $(x+1)(x^2-x+1) = x^3 - x^2 + x + x^2 - x + 1 = x^3 + 1$. Almost everything cancels: this is the "sum of cubes" product.

2.

(a) $51^2 = (50+1)^2 = 2500 + 100 + 1 = 2601$

(b) $49^2 = (50-1)^2 = 2500 - 100 + 1 = 2401$

(c) $103 \cdot 97 = (100+3)(100-3) = 10000 - 9 = 9991$

(d) $999^2 = (1000-1)^2 = 1000000 - 2000 + 1 = 998001$

3.

(a) $6x^2 - 9x = 3x(2x-3)$

(b) $4a^2b - 10ab^2 = 2ab(2a-5b)$

(c) The common factor is $(x-3)$: $x(x-3) + 5(x-3) = (x-3)(x+5)$

4.

(a) Product 12, sum 7: 3 and 4. So $(x+3)(x+4)$.

(b) Product -15, sum -2: -5 and 3. So $(x-5)(x+3)$.

(c) Product 24, sum -11: -3 and -8. So $(x-3)(x-8)$.

(d) Product -56, sum 1: 8 and -7. So $(x+8)(x-7)$.

5.

(a) $x^2 - 81 = (x+9)(x-9)$

(b) $4x^2 - 25y^2 = (2x)^2 - (5y)^2 = (2x+5y)(2x-5y)$

(c) $3x^2 - 48 = 3(x^2 - 16) = 3(x+4)(x-4)$

(d) $x^4 - 16 = (x^2)^2 - 4^2 = (x^2+4)(x^2-4) = (x^2+4)(x+2)(x-2)$. The factor x^2+4 cannot be factorised further.

6.

(a) $ac = 6$, sum 7: 6 and 1. $2x^2 + 6x + x + 3 = 2x(x+3) + (x+3) = (2x+1)(x+3)$

(b) $ac = -12$, sum -1: -4 and 3. $6x^2 - 4x + 3x - 2 = 2x(3x-2) + (3x-2) = (2x+1)(3x-2)$

(c) $ac = 24$, sum -10: -6 and -4. $3x^2 - 6x - 4x + 8 = 3x(x-2) - 4(x-2) = (3x-4)(x-2)$

7.

(a) $x-4=0$ or $2x+3=0$, so $x=4$ or $x=-\frac{3}{2}$.

(b) $x^2-7x=0 \Rightarrow x(x-7)=0 \Rightarrow x=0$ or $x=7$. (Do not divide by x: you would lose $x=0$.)

(c) $(x-8)(x+3)=0 \Rightarrow x=8$ or $x=-3$.

(d) $(2x-3)(2x+3)=0 \Rightarrow x=\frac{3}{2}$ or $x=-\frac{3}{2}$. (Or: $x^2=\frac{9}{4}$.)

8.

(a) $x=7$ or $x=-7$.

(b) $x-3=\pm\sqrt{10}$, so $x=3+\sqrt{10}$ or $x=3-\sqrt{10}$.

(c) $2(x+1)^2=18 \Rightarrow (x+1)^2=9 \Rightarrow x+1=\pm 3 \Rightarrow x=2$ or $x=-4$.

(d) $x^2=-4$. A square is never negative, so there are no solutions.

9.

(a) Half of 6 is 3: $x^2+6x-4=(x+3)^2-9-4=(x+3)^2-13$. So $p=3$, $q=-13$.

(b) $x^2-8x+5=(x-4)^2-16+5=(x-4)^2-11$. So $(x-4)^2=11$, $x=4\pm\sqrt{11}$.

(c) $2x^2+12x+7=2(x^2+6x)+7=2\big((x+3)^2-9\big)+7=2(x+3)^2-11$.

10.

(a) $a=2$, $b=-3$, $c=-4$. $D=9+32=41>0$: two solutions.

$$x=\frac{3\pm\sqrt{41}}{4}$$

(b) $D=16-16=0$: one solution, $x=\frac{-4}{2}=-2$. (Indeed $x^2+4x+4=(x+2)^2$.)

(c) $D=1-24=-23<0$: no solutions.

(d) $D=4+4=8>0$: two solutions, $x=\frac{2\pm\sqrt{8}}{2}=\frac{2\pm 2\sqrt{2}}{2}=1\pm\sqrt{2}$.

11.

(a) $a=-2<0$, so it is a *bergparabool* (opens downwards).

(b) $x_{\text{top}}=-\frac{8}{2\cdot(-2)}=2$, $y_{\text{top}}=-8+16+10=18$. Vertex $(2,18)$, axis of symmetry $x=2$. Check by completing the square: $-2(x^2-4x)+10=-2\big((x-2)^2-4\big)+10=-2(x-2)^2+18$.

(c) y-axis: $x=0$ gives $(0,10)$. x-axis: $-2x^2+8x+10=0 \Rightarrow x^2-4x-5=0 \Rightarrow (x-5)(x+1)=0$, so $(5,0)$ and $(-1,0)$. Note that $x=2$ lies exactly halfway between -1 and 5.

(d) A downward parabola through $(-1,0)$, $(0,10)$, top $(2,18)$, $(4,10)$ (mirror image of $(0,10)$) and $(5,0)$.

(e) $-2x^2+8x+10=2x+10 \Rightarrow -2x^2+6x=0 \Rightarrow -2x(x-3)=0 \Rightarrow x=0$ or $x=3$. Points $(0,10)$ and $(3,16)$.

12.

(a) $2x+1=3(x-3)=3x-9 \Rightarrow x=10$. Check: $x-3=7\neq 0$. Solution $x=10$.

(b) Multiply by x (with $x\neq 0$): $6=x^2-x \Rightarrow x^2-x-6=0 \Rightarrow (x-3)(x+2)=0$. So $x=3$ or $x=-2$; neither is 0, so both are solutions.

(c) Same denominators, so $x^2=9$ with $x\neq 3$. Then $x=3$ or $x=-3$, but $x=3$ makes the denominator zero. Only solution: $x=-3$.

(d) Form $\frac{A}{B} = \frac{A}{C}$: $x^2-4=0$ or $x+1=2x-3$. So $x=2$, $x=-2$ or $x=4$. Check denominators: at $x=2$: 3 and 1; at $x=-2$: -1 and -7; at $x=4$: 5 and 5. None are zero, so all three are solutions.

13.

(a) $x^2-x-6=0 \Rightarrow (x-3)(x+2)=0 \Rightarrow x=3$ or $x=-2$. *Dalparabool*, below the axis between the roots: $-2<x<3$.

(b) $2x^2-5x-3\geq 0$. Roots: $(2x+1)(x-3)=0 \Rightarrow x=-\frac{1}{2}$ or $x=3$. *Dalparabool*, on or above the axis outside the roots: $x\leq -\frac{1}{2}$ or $x\geq 3$.

(c) $D=4-20=-16<0$ and $a>0$, so the parabola lies entirely above the x-axis. The inequality holds for **all** x. (Or: $x^2+2x+5=(x+1)^2+4\geq 4>0$.)

School-exam level

14.

(a) At $t=0$: $h=25$. The roof is 25 m high.

(b) $t_{\text{top}}=-\frac{20}{2\cdot(-5)}=2$, $h=-20+40+25=45$. Maximum height 45 m after 2 seconds.

(c) $-5t^2+20t+25=0 \Rightarrow t^2-4t-5=0 \Rightarrow (t-5)(t+1)=0$. Since $t\geq 0$: after 5 seconds.

(d) $-5t^2+20t+25>40 \Rightarrow -5t^2+20t-15>0 \Rightarrow t^2-4t+3<0 \Rightarrow (t-1)(t-3)<0$. So $1<t<3$: the ball is above 40 m for 2 seconds.

15.

(a) Two sides of x m use $2x$ m, so the side parallel to the wall is $60-2x$ m. Area $A=x(60-2x)$.

(b) $A=-2x^2+60x$, a *bergparabool*. $x_{\text{top}}=-\frac{60}{2\cdot(-2)}=15$, $A=15\cdot 30=450$. Maximum area 450 m^2, with sides 15 m (perpendicular to the wall) and 30 m (along the wall).

(c) $x(60-2x)=400 \Rightarrow -2x^2+60x-400=0 \Rightarrow x^2-30x+200=0 \Rightarrow (x-10)(x-20)=0$. So $x=10$ (enclosure 10 by 40) or $x=20$ (enclosure 20 by 20).

16. The outer rectangle measures $(12+2x)$ by $(8+2x)$.

$$(12+2x)(8+2x)=140 \Rightarrow 96+40x+4x^2=140 \Rightarrow 4x^2+40x-44=0 \Rightarrow x^2+10x-11=0$$

$(x+11)(x-1)=0$, and a width must be positive, so the path is 1 m wide.

17. Let the numbers be n and $n+2$ with n odd. $n(n+2)=255 \Rightarrow n^2+2n-255=0 \Rightarrow (n-15)(n+17)=0$. So $n=15$ or $n=-17$. The pairs are $15, 17$ and $-17, -15$.

18.

(a) $D=p^2-36=0 \Rightarrow p=6$ or $p=-6$.

(b) $D<0 \Rightarrow p^2<36 \Rightarrow -6<p<6$.

(c) $x^2-3x+5=x+p \Rightarrow x^2-4x+5-p=0$. Touching means exactly one intersection: $D=16-4(5-p)=4p-4=0$, so $p=1$. Then $x^2-4x+4=0$, $x=2$, $y=2+1=3$. Point of contact $(2,3)$.

19.

(a) After n increases the price is $10+0.5n$ euros and $300-10n$ tickets are sold. Revenue = price times number: $R=(10+0.5n)(300-10n)$.

(b) $R = 3000 - 100n + 150n - 5n^2 = -5n^2 + 50n + 3000$. $n_{\text{top}} = -\frac{50}{2\cdot(-5)} = 5$. Price $10 + 2.50 =$ €12.50, tickets $300 - 50 = 250$, revenue $12.50 \cdot 250 =$ €3125.

20.

(a) Time = distance / speed. At speed v the trip takes $\frac{60}{v}$ hours; at $v+5$ it takes $\frac{60}{v+5}$ hours. The difference is 1 hour.

(b) Multiply by $v(v+5)$ (both non-zero since $v > 0$):

$$60(v+5) - 60v = v(v+5) \Rightarrow 300 = v^2 + 5v \Rightarrow v^2 + 5v - 300 = 0 \Rightarrow (v-15)(v+20) = 0$$

Speed must be positive, so $v = 15$ km/h. Check: 4 hours versus 3 hours.

21.

(a) The roots are 1 and 5, and the coefficient of x^2 is 1, so $y = (x-1)(x-5) = x^2 - 6x + 5$: $b = -6$, $c = 5$. The vertex lies halfway between the roots at $x = 3$: $y = (2)(-2) = -4$. Vertex $(3, -4)$.

(b) Vertex form: $y = a(x-2)^2 - 1$. Through $(0, 3)$: $3 = 4a - 1 \Rightarrow a = 1$. So $y = (x-2)^2 - 1 = x^2 - 4x + 3$.

22. Area $w(w+3)$ with $w > 0$.

- At least 40: $w^2 + 3w - 40 \geq 0 \Rightarrow (w+8)(w-5) \geq 0 \Rightarrow w \leq -8$ or $w \geq 5$. With $w > 0$: $w \geq 5$.
- At most 70: $w^2 + 3w - 70 \leq 0 \Rightarrow (w+10)(w-7) \leq 0 \Rightarrow -10 \leq w \leq 7$.

Both together: $5 \leq w \leq 7$ (width between 5 and 7 cm).

Challenge

23.

(a) $(x^2+2)^2 = x^4 + 4x^2 + 4$, so $x^4 + 4 = (x^2+2)^2 - 4x^2 = (x^2+2)^2 - (2x)^2$. Difference of squares:

$$x^4 + 4 = (x^2 + 2 + 2x)(x^2 + 2 - 2x) = (x^2 + 2x + 2)(x^2 - 2x + 2)$$

(b) By (a), $n^4 + 4 = (n^2 + 2n + 2)(n^2 - 2n + 2)$. The second factor is $(n-1)^2 + 1$.

- For $n = 1$: $n^4 + 4 = 5$, which is prime.
- For $n \geq 2$: $(n-1)^2 + 1 \geq 2$, and the first factor $n^2 + 2n + 2$ is larger still. So $n^4 + 4$ is a product of two whole numbers that are both at least 2, hence not prime.

So $n = 1$ is the only one. (Check: $2^4 + 4 = 20$, $3^4 + 4 = 85 = 5 \cdot 17$, $4^4 + 4 = 260$.)

24. Suppose $n^2 + 60 = m^2$ with m a positive integer. Then $m^2 - n^2 = 60$, so $(m-n)(m+n) = 60$ with $0 < m - n < m + n$.

The factors $m - n$ and $m + n$ differ by $2n$, an even number, so they have the same parity. Their product 60 is even, so they cannot both be odd: both are even. The factor pairs of 60 with both factors even and the first smaller are $2 \cdot 30$ and $6 \cdot 10$.

- $m - n = 2$, $m + n = 30$: $m = 16$, $n = 14$. Check: $196 + 60 = 256 = 16^2$.
- $m - n = 6$, $m + n = 10$: $m = 8$, $n = 2$. Check: $4 + 60 = 64 = 8^2$.

So $n = 2$ or $n = 14$.

25. Substitute $u = x^2 - 3x$: $u^2 - 2u - 8 = 0 \Rightarrow (u-4)(u+2) = 0$, so $u = 4$ or $u = -2$.

- $x^2 - 3x = 4 \Rightarrow x^2 - 3x - 4 = 0 \Rightarrow (x-4)(x+1) = 0 \Rightarrow x = 4$ or $x = -1$.
- $x^2 - 3x = -2 \Rightarrow x^2 - 3x + 2 = 0 \Rightarrow (x-1)(x-2) = 0 \Rightarrow x = 1$ or $x = 2$.

Solutions: $x = -1,\ 1,\ 2,\ 4$. Expanding first would give a degree-4 equation; the substitution keeps it quadratic.

26.

(a) Expand: $x^2 - (a+b)x + ab - k^2 = 0$. The discriminant is

$$D = (a+b)^2 - 4(ab - k^2) = a^2 + 2ab + b^2 - 4ab + 4k^2 = (a-b)^2 + 4k^2$$

A sum of two squares is never negative, so $D \geq 0$ and there is at least one real solution. (There is exactly one only when $a = b$ and $k = 0$.)

Alternative without the discriminant: the parabola $y = (x-a)(x-b)$ is a *dalparabool* through $(a, 0)$, so its lowest value is at most $0 \leq k^2$, and it grows without bound. So it must reach the height k^2 somewhere.

(b) Since $\sqrt{p} - \sqrt{q}$ is a real number, $(\sqrt{p} - \sqrt{q})^2 \geq 0$, so $p - 2\sqrt{pq} + q \geq 0$, i.e. $\frac{p+q}{2} \geq \sqrt{pq}$. Equality holds exactly when $\sqrt{p} - \sqrt{q} = 0$, that is $p = q$.

For a rectangle with sides p and q and perimeter P: $p + q = \frac{P}{2}$. Then the area satisfies $pq \leq \left(\frac{p+q}{2}\right)^2 = \frac{P^2}{16}$, with equality only when $p = q$. So the square with side $\frac{P}{4}$ has the largest area.

27. By Vieta: $x_1 + x_2 = 5$ and $x_1 x_2 = 3$.

(a) $x_1^2 + x_2^2 = (x_1 + x_2)^2 - 2x_1x_2 = 25 - 6 = 19$

(b) $\frac{1}{x_1} + \frac{1}{x_2} = \frac{x_1 + x_2}{x_1 x_2} = \frac{5}{3}$

(c) $(x_1 - x_2)^2 = (x_1 + x_2)^2 - 4x_1x_2 = 25 - 12 = 13$. (This is exactly the discriminant: $D = b^2 - 4ac$ with $a = 1$.)

(d) $(x_1+x_2)^3 = x_1^3 + 3x_1^2x_2 + 3x_1x_2^2 + x_2^3 = x_1^3 + x_2^3 + 3x_1x_2(x_1+x_2)$, so $x_1^3 + x_2^3 = 125 - 3\cdot 3\cdot 5 = 80$.

28. For the roots x_1, x_2 of $x^2 - 4x + 1 = 0$: $x_1 + x_2 = 4$, $x_1 x_2 = 1$. The new roots x_1^2 and x_2^2 have

- sum $x_1^2 + x_2^2 = 16 - 2 = 14$,
- product $x_1^2 x_2^2 = (x_1 x_2)^2 = 1$.

So the equation is $x^2 - 14x + 1 = 0$. (Check: the roots of $x^2 - 4x + 1$ are $2 \pm \sqrt{3}$, and $(2 \pm \sqrt{3})^2 = 7 \pm 4\sqrt{3}$, which add up to 14 and multiply to $49 - 48 = 1$.)

29.

(a) Square: $\left(x + \frac{1}{x}\right)^2 = x^2 + 2 + \frac{1}{x^2} = 9$, so $x^2 + \frac{1}{x^2} = 7$. Cube: $\left(x + \frac{1}{x}\right)^3 = x^3 + 3x + \frac{3}{x} + \frac{1}{x^3} = x^3 + \frac{1}{x^3} + 3\left(x + \frac{1}{x}\right)$, so $27 = x^3 + \frac{1}{x^3} + 9$ and $x^3 + \frac{1}{x^3} = 18$.

(b) Multiply by x ($x \neq 0$): $x^2 + 1 = 3x \Rightarrow x^2 - 3x + 1 = 0 \Rightarrow x = \frac{3 \pm \sqrt{5}}{2}$. Neither root is 0. Check with $x = \frac{3+\sqrt{5}}{2}$: then $\frac{1}{x} = \frac{2}{3+\sqrt{5}} = \frac{2(3-\sqrt{5})}{9-5} = \frac{3-\sqrt{5}}{2}$ (the two roots multiply to 1, by Vieta). Now $x^2 = \frac{14+6\sqrt{5}}{4}$ and $\frac{1}{x^2} = \frac{14-6\sqrt{5}}{4}$, which add up to $\frac{28}{4} = 7$. Correct.

30.

(a) Two cases.

- $m = 1$: the x^2-term disappears and the equation is linear: $-2x + 3 = 0$, so $x = \frac{3}{2}$. Exactly one solution.
- $m \neq 1$: a quadratic equation with $D = 4m^2 - 4(m-1)(m+2) = 4m^2 - 4(m^2 + m - 2) = -4m + 8$. Exactly one solution when $D = 0$, so $m = 2$. Then $x^2 - 4x + 4 = 0$, $x = 2$.

Answer: $m = 1$ or $m = 2$. The trap is forgetting that the abc-formula only applies when the coefficient of x^2 is not zero.

(b) $x^2 + 1 = mx \Rightarrow x^2 - mx + 1 = 0$. Touching: $D = m^2 - 4 = 0$, so $m = 2$ or $m = -2$.

- $m = 2$: $x^2 - 2x + 1 = 0$, $x = 1$, point $(1, 2)$.
- $m = -2$: $x^2 + 2x + 1 = 0$, $x = -1$, point $(-1, 2)$.

The two tangent lines are mirror images in the y-axis, as you would expect from the symmetry of $y = x^2 + 1$.